

DeepSense 6G Scenario 36 — 25-Cell EDA & Data Audit Guide

Comprehensive Walkthrough of Exploratory Data Analysis, Data Leakage Audit, Label Arithmetic, and Weak-Link Filtering

EDA Objective & Scope: This document provides an exhaustive breakdown of the 25-cell Exploratory Data Analysis conducted on scenario36.csv and the 99,196 raw power files. It documents key empirical findings including linear power units, overall-beam label construction, antenna panel dominance, polar bearing geometry, the discovery of the **0% Train/Test location-cluster leakage bug** under naive splitting, camera complementarity, and the derivation of the **6.0 dB Weak-Link Threshold**.

1. Data Ingestion, Structure & Label Verification (Cells 1–4)

Initial verification established the physical data layout and label construction rules across the dataset's 24,799 rows:

- **Cell 1 (First Look):** Loaded `scenario36.csv` without prior assumptions, inspecting shape, data types, and fully expanded columns.
- **Cell 2 (Path Resolution):** Verified file paths under `scenario36/`, confirming structure for `unit1/` power files and camera images.
- **Cell 3 (Raw Power Inspection & _v2 Column Audit):** Inspected raw `.txt` power files. Confirmed power values are **linear power (mW)**, NOT dB! Verified that `\_v2` summary columns are 100% identical to original columns across all rows.
- **Cell 4 (Label Arithmetic Verification):** Confirmed exact label construction logic:
 - `unit1\_pwr1\_best-beam` = `argmax` of 64-value beam file.
 - `unit1\_overall-beam` = `argmax` of 4 concatenated array files (\$4 \times 64 = 256\$ total beams).

2. Spatial Antenna Imbalance & Bearing Geometry (Cells 5–7)

Empirical analysis revealed a strong, non-uniform spatial distribution across the 4 receiver antenna panels:

Antenna Panel	Beam Index Range	Dominance Frequency	Dominant Compass Bearing (Cell 7)
Panel 1 (pwr1)	Beams 0 – 63	14,582 rows (58.8%)	East / South-East (90° – 150°)
Panel 2 (pwr2)	Beams 64 – 127	1,314 rows (5.3%)	South / South-West (180° – 240°)
Panel 3 (pwr3)	Beams 128 – 191	8,456 rows (34.1%)	North / North-West (300° – 360°)
Panel 4 (pwr4)	Beams 192 – 255	447 rows (1.8%)	West (240° – 300°)

- **Geometric Proof (Cells 6 & 7):** Computing relative bearing $\theta = \text{atan2}(\Delta \text{lon}, \Delta \text{lat})$ and plotting polar histograms proved that panel selection is driven strictly by vehicle relative geometry rather than random noise.

3. Trajectory Analysis & The Spatial Leakage Bug (Cells 8–11)

Trajectory analysis revealed 124 distinct sequential driving runs (`seq_index`). Mapping chronological order against physical lat/lon clusters uncovered a critical evaluation flaw in standard naive splitting:

The Spatial Leakage Bug (Cell 11):

Under a standard naive chronological split (e.g. 55% train, 15% val, 15% calib, 15% test):

- Total Location Clusters in Train: **14 clusters**
- Total Location Clusters in Test: **8 clusters**
- Clusters in Test **ALSO** appearing in Train: **0 clusters (0%)!**
- Test Clusters **NEVER** seen in Train: **8 out of 8 (100%)!**

• **Impact & Solution:** Because naive chronological splitting forces the test set onto completely unseen street segments, models trained without Drive-Block trajectory splitting experience severe spatial collapse. Drive-Block splitting is essential for honest out-of-distribution evaluation.

4. Camera Views & Environmental Quality Audit (Cells 12–16)

- **Dual Camera Complementarity (Cells 12 & 13):** Compared `rgb5`` and `rgb6`` wide-angle front-facing cameras. Proved both carry non-redundant visual context (sometimes `rgb5`` captures target vehicle better, sometimes `rgb6``).
- **Lighting Invariance (Cell 14):** Timestamp analysis proved 100% of rows were recorded between 11:00–14:00 (uniform daytime lighting, eliminating day/night domain shift concerns).
- **GPS Fix Audit (Cells 15 & 16):** Analyzed DOP parameters. Identified 7 degraded HDOP > 2 rows in Unit 2; verified coordinates remain physically smooth and valid.

5. Feasibility Gate & Weak-Link Filter Derivation (Cells 17–23)

Cell 17 evaluated all 256-beam power profiles across 24,799 rows (99,196 power files), explicitly converting linear power to dB scale ($10 \log_{10}(P + 1e-12)$):

- **Near-Optimal Set Size:** At $\Delta = 3.0 \text{ dB}$, median near-optimal set size is sharp (3–5 beams), confirming beam tracking feasibility.
- **Degenerate Outliers (Cell 18):** Identified flat profile outliers where up to 256 beams fell within 3 dB of optimal.
- **Dynamic Range Metric (Cells 19 & 20):** Defined $\text{range}_{\text{dB}} = P_{\text{max, dB}} - P_{\text{min, dB}}$. Binned dynamic range finely to locate the elbow transition point.

The Weak-Link Threshold (Cell 21):

Setting `WEAK_LINK_THRESHOLD_DB = 6.0` dB flags **2,763 rows (11.14%)** as blocked/flat profiles. Physical mapping (Cells 22 & 23) confirmed weak-link rows cluster along a rural highway stretch where inter-vehicle distance reaches **229 meters** (vs 8 meters for strong signals)—representing physical path loss rather than sensor defects.

6. Data Cleanliness Audit Summary (Cells 24 & 25)

- **Duplicate Rows:** 0 fully duplicated rows (0.0%).
- **Primary Key Integrity:** 0 duplicated `abs_index`` values.
- **Missing Values:** Exactly 7 missing values total in `unit2_gps1_altitude``, matching the exact 7 HDOP dip rows from Cell 15.